



D5 Natural selection and evolution

D · Continuity and change · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Evolution is the change in allele frequencies in a population over successive generations. Natural selection requires heritable variation in a population (arising from mutation and, in sexually reproducing organisms, recombination); when this variation affects survival or reproductive success in a given environment, individuals with advantageous traits tend to survive and reproduce more, passing their alleles on more frequently, gradually shifting the population's allele frequencies. Evidence for evolution includes the fossil record, comparative anatomy (homologous structures reflecting common ancestry), molecular similarities in DNA/protein sequences, and directly observed examples such as the spread of antibiotic resistance in bacteria.

Worked example: follow the reasoning

Modelling the spread of antibiotic resistance in a bacterial population. Suppose a population starts with 2% of cells carrying a resistance allele (from a random mutation) before antibiotic exposure. Step 1: when the antibiotic is applied, susceptible bacteria (98%) are killed or fail to reproduce, while resistant bacteria survive. Step 2: surviving resistant bacteria reproduce, and in the next generation the proportion carrying the resistance allele might rise sharply, e.g. from 2% to 40%, since non-resistant competitors have been largely removed. Step 3: with continued antibiotic use, resistant bacteria continue to be favoured, and the resistant allele's frequency approaches 100% over further generations. Conclusion: this rapid rise in allele frequency, driven purely by differential survival and reproduction under a strong selective pressure (the antibiotic), is a direct, observable example of natural selection acting on a population.

Quick check · multiple choice

Which observation best supports the idea that natural selection, rather than the environment directly causing new traits, drives evolutionary change?

A Mutations arise randomly, independent of any need, but the environment determines which existing variants are favoured

B All individuals in a population develop identical adaptations when exposed to the same environment

C Environmental exposure directly alters an organism's DNA sequence to produce a needed trait

D Traits acquired during an organism's lifetime are passed on to its offspring

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A population of beetles shows variation in shell colour from very pale to very dark. Before a change in habitat vegetation, most beetles were pale-coloured, matching pale sandy ground. After the vegetation changed to darker soil and leaf litter, researchers recorded shell colour in the beetle population over five generations and found the proportion of dark-coloured beetles increased substantially, while pale-coloured beetles became rare. [7]
- (a) Using the theory of natural selection, explain the change observed in the beetle population over the five generations. [4]
- (b) Explain why heritability of shell colour is essential for this change in the population to occur through natural selection. [2]
- (c) State one type of evidence, other than direct field observation, that could further support the occurrence of natural selection in this population. [1]

Final answer: _____



Answers and explanation

Question 1 [7]

(a) Shell colour varies within the beetle population due to heritable genetic variation; on darker soil, predators can more easily spot and eat pale beetles (which are now poorly camouflaged) than dark beetles (which are now well camouflaged); as a result, dark beetles survive and reproduce more successfully than pale beetles, passing on alleles for dark colour more frequently; over successive generations, this differential survival and reproduction increases the proportion of alleles for dark colouration in the population, so dark beetles become more common and pale beetles rarer.

(b) The trait must be heritable (genetically determined and passed to offspring) for differential survival/reproduction to change the allele frequencies of the next generation; if shell colour were purely due to environmental influence and not genetically determined, selective survival of dark beetles would not change the genetic composition of future generations.

(c) Any valid example, such as: comparing DNA sequences of genes controlling pigmentation between beetle populations from different habitats to look for changes in allele frequency consistent with selection, or examining museum/historical specimens collected before and after the habitat change to directly compare colour-allele frequencies over time.

Quick check answer

A. Variation (from mutation) arises before and independently of the environmental pressure; selection then favours individuals who already happen to carry advantageous alleles, rather than the environment creating new traits on demand.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Biology HL/SL subject guide.

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